

Design Challenge Report

Self-Biased Sub-1V Bandgap Reference Circuit

1. Specifications

Parameter	Value
Technology	65 nm
Supply voltage V_{DD}	1.2 V
Corners	Temp: -40 to 125 °C; Process: TT, SS, SF, FS, FF
Output voltage V_{REF}	0.8 V
Total bias current	$< 10 \mu\text{A}$
Phase margin	$> 60^\circ$

Table 1: Target specifications for the redesigned self-biased sub-1V BGR

2. Part 1: BGR Core + Startup Circuit

2.1 Topology Recap

Circuit Description

- **Core:** MP1 and MP2 are matched PMOS current sources forcing equal currents into the core branches. The amplifier ensures node DC voltages remain equal.
- **CTAT/PTAT:** Q1/Q2 (ratio $N : 1$) generate ΔV_{EB} . R2 defines the PTAT current, and R1/R3 define the CTAT current.
- **Startup:** MP13/MN6/MN7 jumpstart the circuit. MN7 pulls current out of the gate node when off. As the core turns on, $V_{GS,MN6}$ rises, activating MN6 to turn off the startup path.

2.2 Hand Analysis

Design Equations

1. Current Generation:

$$I_{PTAT} = \frac{V_T \ln N}{R_2}$$

$$I_{CTAT} = \frac{V_{EB,Q1}}{R_3}$$

2. Output Voltage (Sub-1V Target):

$$V_{REF} = R_4 (I_{PTAT} + I_{CTAT}) = R_4 \left(\frac{V_T \ln N}{R_2} + \frac{V_{EB,Q1}}{R_3} \right) < 0.8$$

I chose R_2 to be $164.496 \text{ k}\Omega$ to provide 500 nA PTAT.

3. Temperature Compensation Ratio: At T_0 , the slopes must cancel ($\partial V_{REF}/\partial T = 0$):

$$\frac{R_3}{R_2} = \frac{\ln N \cdot |\partial V_T / \partial T|}{|\partial V_{EB} / \partial T|} \approx \ln N \times 21.6$$

I chose $N=24$

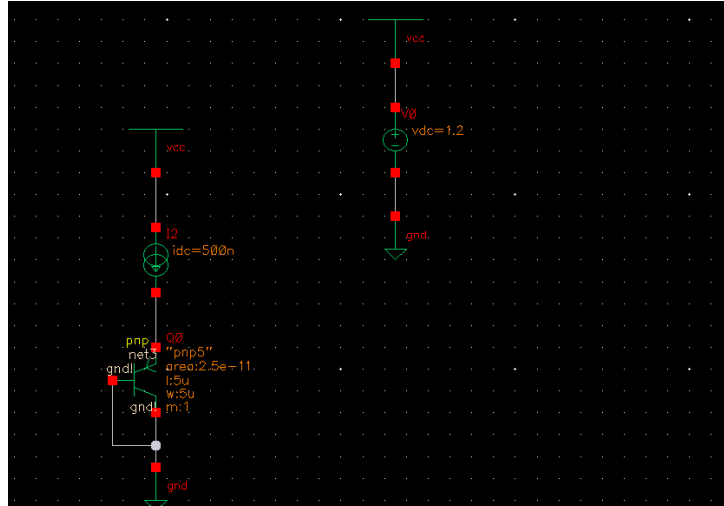
2.3 V_{EB} Temperature Coefficient Characterization

Figure 1: Testbench utilized to characterize the PNP bipolar device, applying a fixed 500 nA collector current to sweep V_{EB} against temperature.

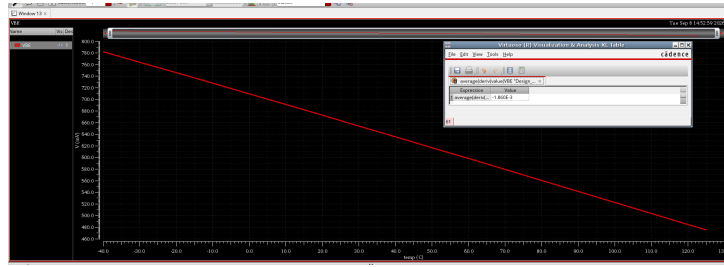


Figure 2: Getting the negative temperature coefficient

2.4 DC Operating Point (Behavioral Amplifier)

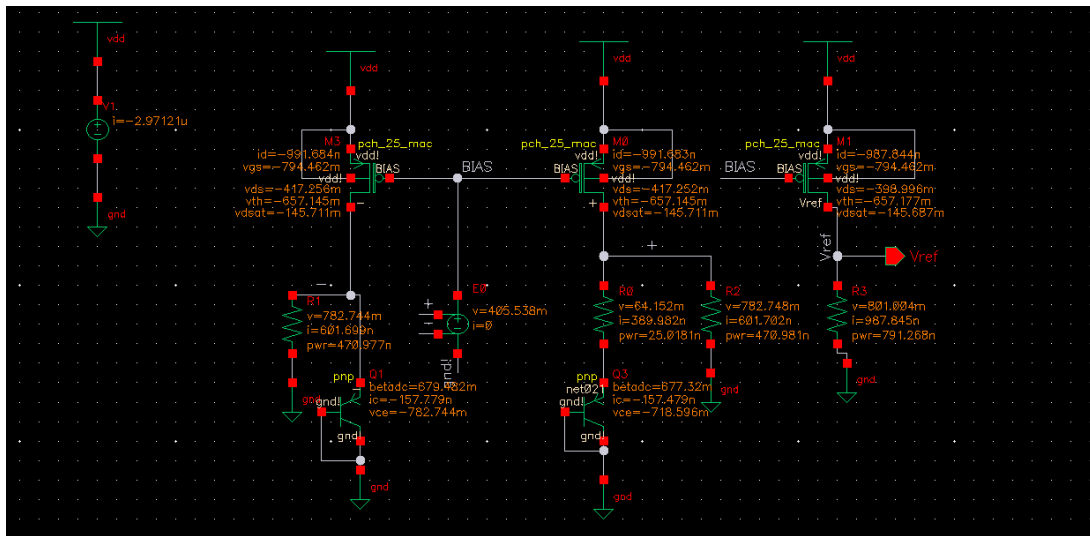


Figure 3: Core + startup circuit with the amplifier replaced by an ideal VCVS (E_0). The DC operating point is annotated. The reference voltage V_{REF} sits at 801 mV (measured across R3), and the bias node is driven to roughly 405.5 mV.

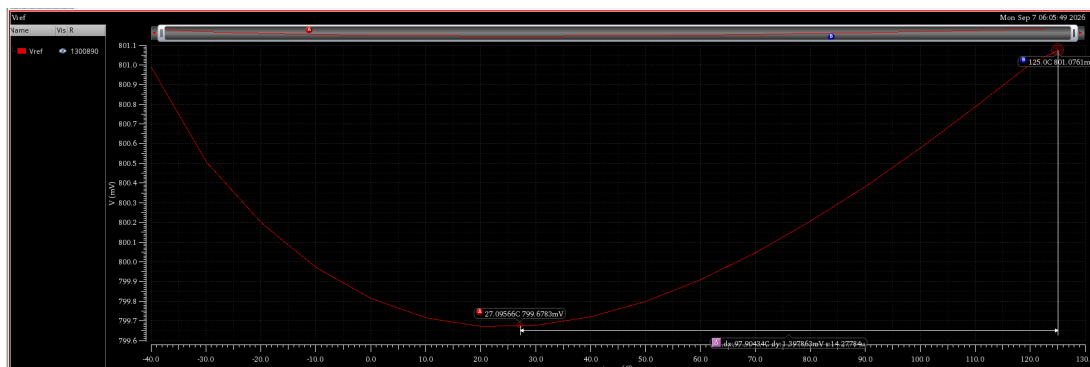


Figure 4: Vref variation

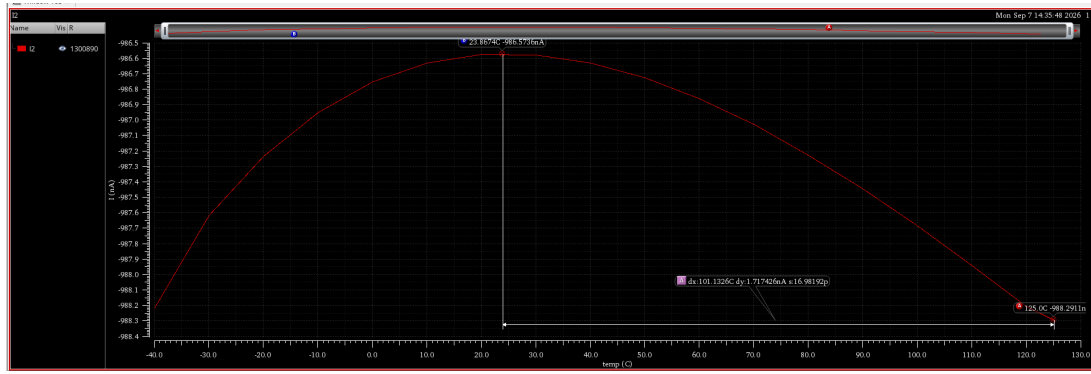


Figure 5: ZTAT current

2.5 V_{out} vs. Temperature – All Corners

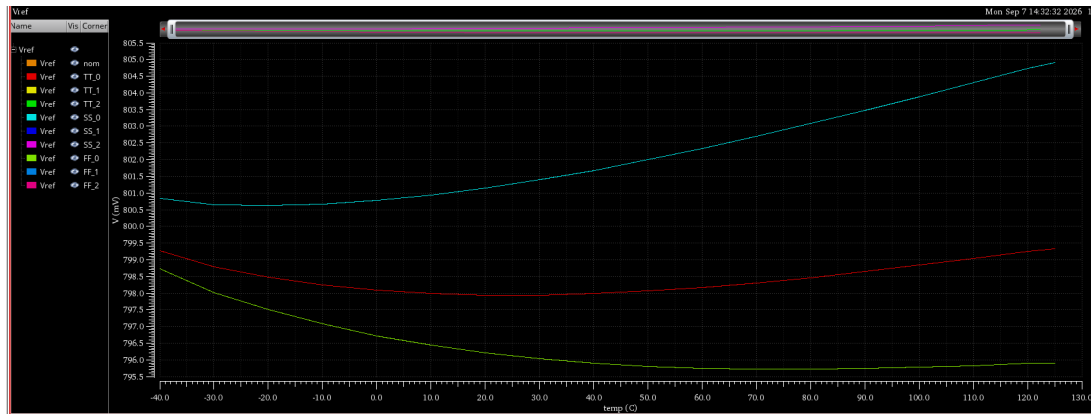


Figure 6: V_{REF} vs. temperature (-40 to 125 °C) for the behavioral-amplifier core. The traces demonstrate a variation ranging from roughly 795.5 mV to 805.5 mV.

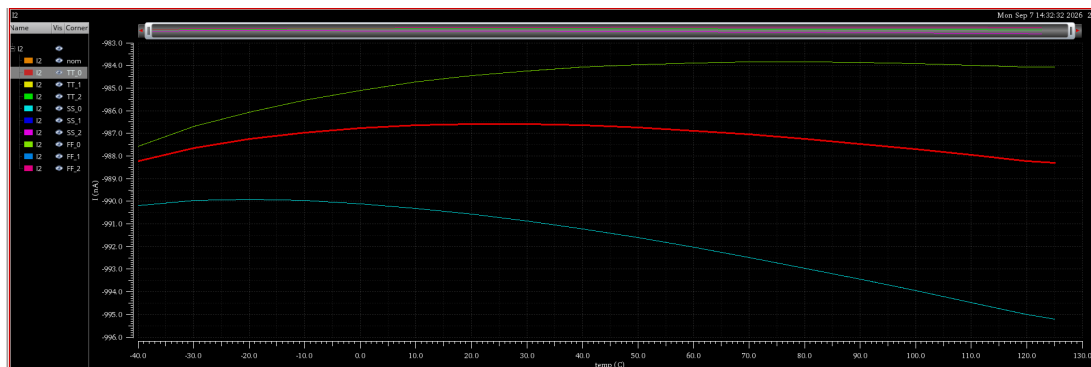


Figure 7: I_{ZTAT} vs. temperature (-40 to 125 °C) for the behavioral-amplifier core.

Parameter	Nominal							TT_0	TT_1	TT_2	SS_0	SS_1	SS_2	FF_0	FF_1	FF_2
cm65gplus... tt_naz5								tt_bip	tt_bip	tt_bip	ss_bip	ss_bip	ss_bip	ff_bip	ff_bip	ff_bip
cm65gplus... tt_rfnvar_25								TT_25	TT_25	TT_25	SS_25	SS_25	SS_25	FF_25	FF_25	FF_25
cm65gplus... tt_dio_naz5...								nom	nom	nom	nom	nom	nom	nom	nom	nom
cm65gplus... tt_25ud18								nom	nom	nom	nom	nom	nom	nom	nom	nom
cm65gplus... tt_rfnos								nom	nom	nom	nom	nom	nom	nom	nom	nom
cm65gplus... tt_25								nom	nom	nom	nom	nom	nom	nom	nom	nom
cm65gplus... tt_dio_25ud...								nom	nom	nom	nom	nom	nom	nom	nom	nom
cm65gplus... tt_dio_25								nom	nom	nom	nom	nom	nom	nom	nom	nom
cm65gplus... tt_mos_cap								nom	nom	nom	nom	nom	nom	nom	nom	nom
cm65gplus... tt								nom	nom	nom	nom	nom	nom	nom	nom	nom
Test	Output	Nominal	Spec	Weight	Pass/Fail	Min	Max	TT_0	TT_1	TT_2	SS_0	SS_1	SS_2	FF_0	FF_1	FF_2
test_luctest1	Vref	798m				796m	805m	799.3m	798m	799.4m	800.9m	801.3m	805m	798.7m	796.2m	796m
test_luctest1	/M1/D															

Figure 8: V_{REF} pass/fail summary for the behavioral OTA design. The table indicates a nominal V_{REF} of 798 mV, with a minimum of 796.2 mV and a maximum of 805 mV across the FF process corners.

3. Part 2: BGR with Actual Self-Biased Amplifier

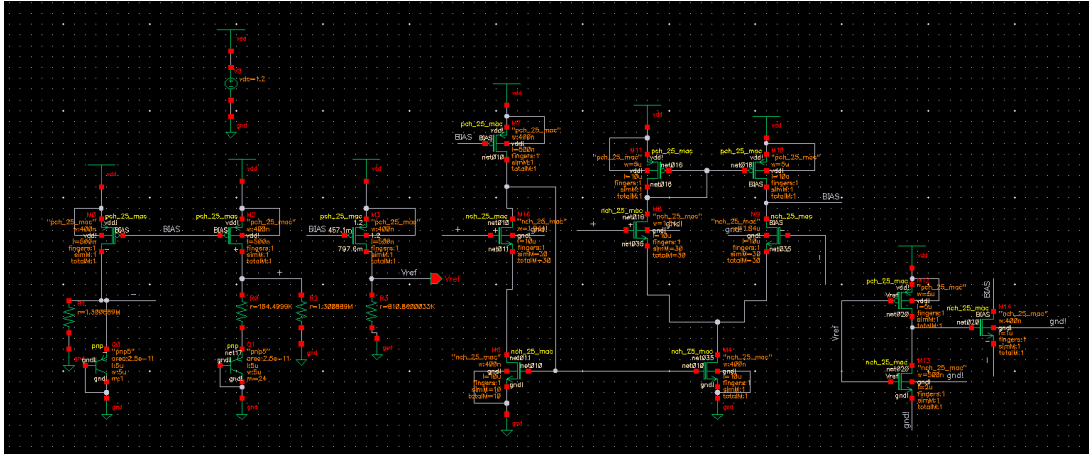


Figure 9: Sizing of the BGR after fine tuning to keep all devices in saturation and keeping $V_+ = V_-$ by using ADT knob at maximum $\frac{g_m}{g_{ds}}$

3.1 DC Operating Point

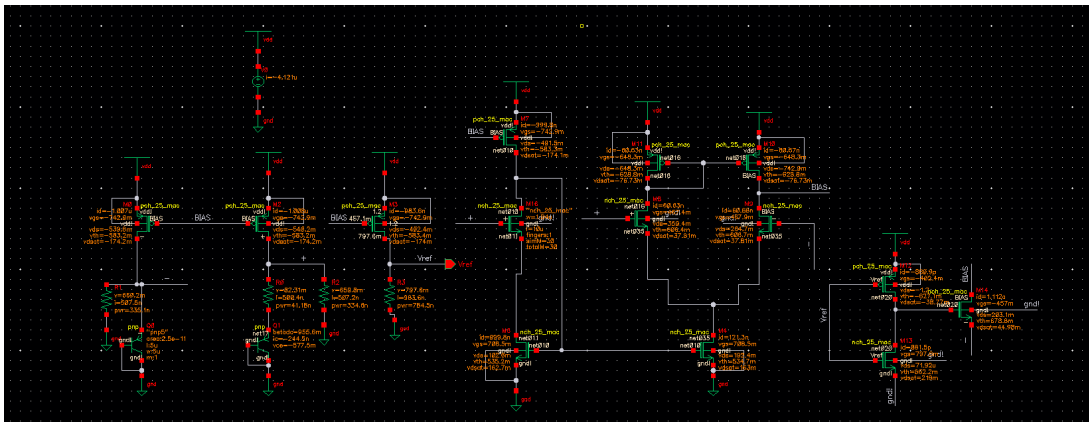


Figure 10: Full self-biased BGR including the actual amplifier and the startup circuit

3.2 Vout vs. Temperature – All Corners

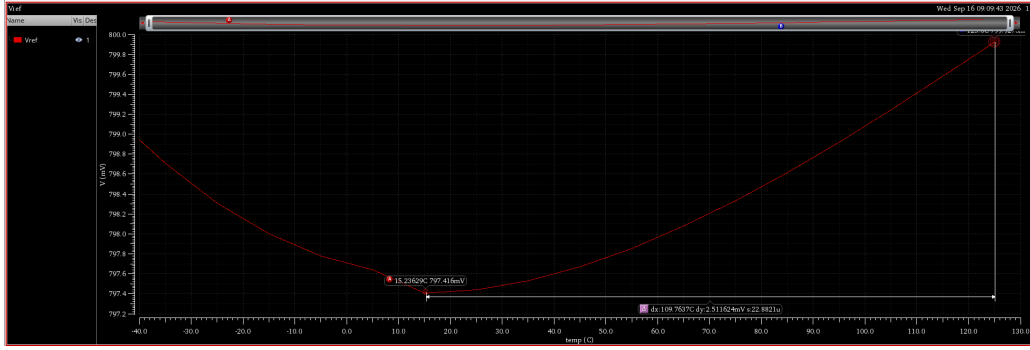


Figure 11: V_{REF} vs. temperature across TT i.e : 2.5 mV

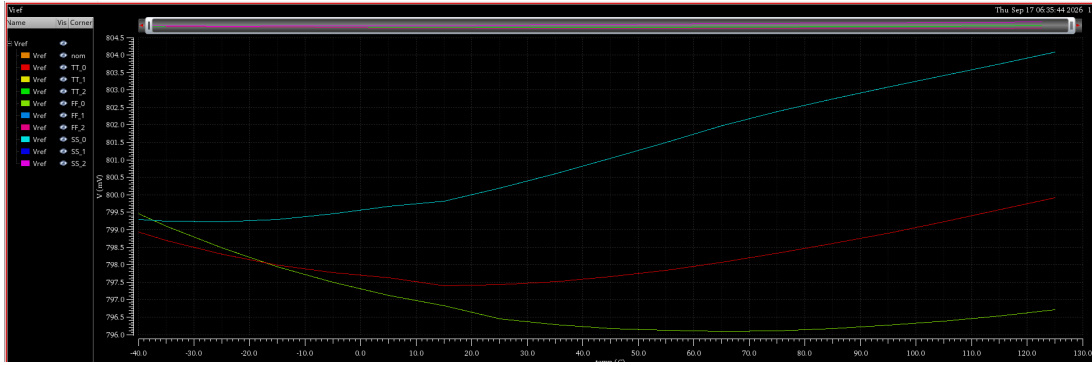


Figure 12: V_{REF} vs. temperature across all process corners using the actual self-biased amplifier

3.3 Stability (STB) Analysis

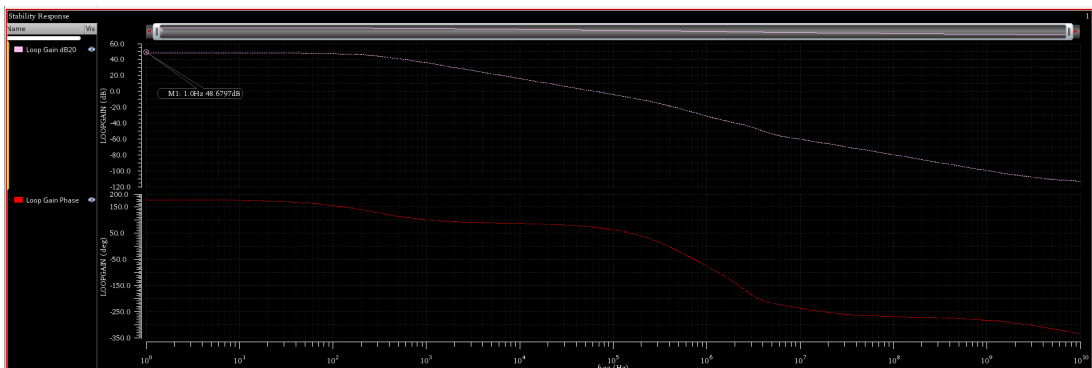


Figure 13: Open-loop gain of the self-biased loop and phase

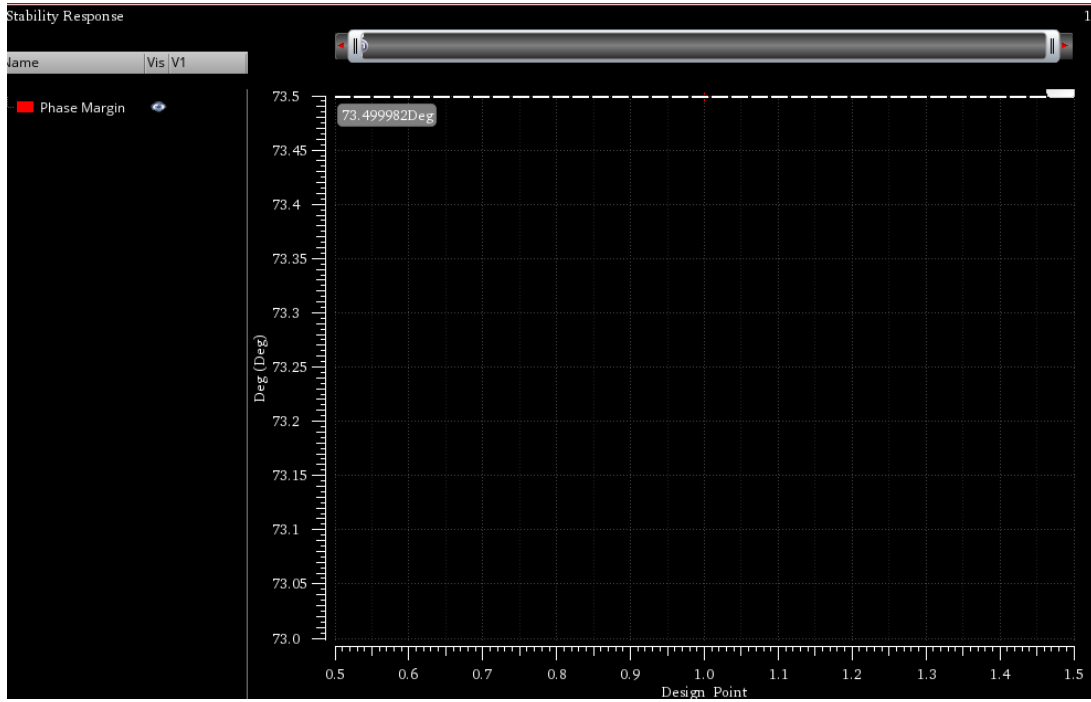


Figure 14: Phase Margin from stb summary

3.4 Transient Simulation (V_{DD} Ramp)

Figure 15: Transient V_{DD} and V_{ref}

Parameter	Target	Achieved
Output voltage V_{REF}	0.8 V	796 mV – 804.5 mV* (Across Corners)
Total bias current	$< 10 \mu\text{A}$	$4.12 \mu\text{A}$
Phase margin	$> 60^\circ$	73.5°

* SF and FS corners were not covered as they were unavailable for the BJT transistors in this PDK.

Table 2: Target vs. Achieved Specifications for the Self-Biased Sub-1V BGR